A natural question that arises in the context of chaotic systems is whether such a trajectory average in fact makes sense. Considering any conservative system, we know that the energy itself is a conserved quantity along trajectories, and its average is therefore a function of the initial condition. On the other hand, imagine trajectories emanating from different points on a single energy level set Σ_{E}, can we necessarily conclude that all such trajectories will yield the same average value for any observable? This question is obviously important as the initial conditions of a molecular system are typically generated through approximations and estimations—it would be disturbing if the choice of these (at uniform energy) were to greatly impact the computation of an average.

One situation in which we would expect different initial conditions to yield different trajectory averages is if the energy surface has several connected components.

Example 5.3 Consider the *double well potential*

$$\phi(q) = (1 - q^2)^2.$$

The energy $H(q, p) = p^2/2 + \phi(q)$ will tend to $+\infty$ as $q \to \infty$ or $p \to \infty$, but is obviously bounded below (by zero). The graph of the energy, along with some level curves, is shown in Fig. 5.2. From the graph it is clear that the energy surface $H(q, p) = \mathrm{E}$ has more than one component if $0 < \mathrm{E} < 1$. If an initial condition is taken from one of the two basins, say with energy E_0, then the trajectory through that point will remain forever in the one basin, thus the averages of many quantities, if computed using the trajectory values, would not be a good approximation of the overall average at constant energy E_0.

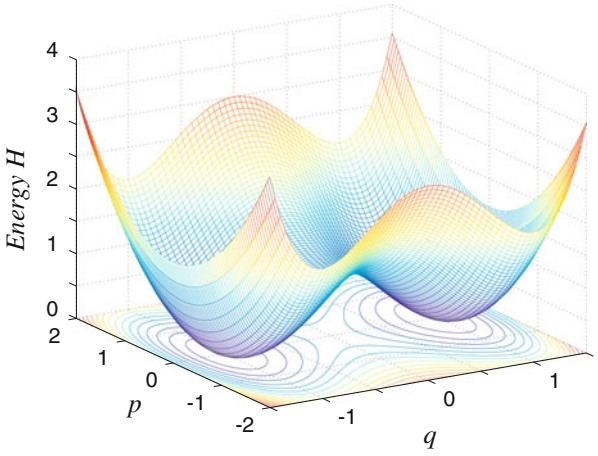


Fig. 5.2 Energy function for the double well. The energy surface corresponding to a value of energy $\mathrm{E} \in (0, 1)$ is not connected

More generally, if a given system has a first integral (conserved quantity) $I = I(z)$, different from the energy, which is preserved along trajectories, then the average of this conserved quantity (and any other quantity that depends upon it) will be dependent on the choice of initial condition.

5.4.1 The Concept of Ergodicity

While being mindful of the examples given above, there are many systems which, at least for a useful range of energy, appear to have the property that averages are independent of initial condition. Systems for which the trajectory averages are the same as phase space averages on the energy surface are said to be *ergodic*.[6] The concept of ergodicity can be introduced in the setting of general dynamical systems, but it is very challenging to analyze this property for realistic systems. An analogous concept for stochastic differential equations will be introduced in the next chapter; in this setting of randomized models ergodicity is much more amenable to mathematical study. Nonetheless, we motivate the concept in this section using deterministic systems and numerical examples.

The formal definition of ergodicity takes one of several forms. For our current purposes, we opt for a definition that is close to our intended goal (the computation of averages).

Definition 5.1 A Hamiltonian system is said to be ergodic on an energy surface Σ_{E} (microcanonically ergodic) if, for almost all trajectories Γ emanating from initial conditions on Σ_{E}, and for any observable g, the following holds

$$\bar{g}_{\Gamma} = \mathrm{Av}_{H=\mathrm{E}}(g),$$

where $\bar{g}_{\Gamma}$ is defined in (5.9). By "almost all" we mean that there could be a low-dimensional subset for which the trajectory averages fail to equal the space averages. The term "almost all" thus allows for the existence of some fixed points and periodic orbits, which are almost certain to be present in a deterministic system.

Alternatively, we may say that a dynamical system as ergodic if the probability measure of any invariant set $\mathcal{A}$ of the flow is either 0 or 1. An invariant set $\mathcal{A}$ of the flow is one such that

$$\mathscr{F}_t(\mathcal{A}) = \mathcal{A},$$

for all $t \neq 0$. To see that this is the same concept of ergodicity as defined above, let χ be the "indicator function" of the complement of $\mathcal{A}$ whose values is zero if $z \in \mathcal{A}$

[6]The term "ergodic" was coined by Boltzmann in 1887. He joined the Greek term *ergon* ("action," "work") with *hodos* ("path") to refer to an energy-path. An ergodic path one that traverses the entire energy surface eventually touching all points thereof.